Software Release Checklist

Open Source Software Project Release

An open source software project release is for the entirety of the project. The default branch (i.e. `master` or `main`) will be used for the open source software project release.

Open source software project releases allow the source code to be made available publicly. This may be in the timely interest of best science, to facilitate external domain and/or technical code reviews (in pursuance of a subsequent scientific information product release), or for any other reason deemed acceptable by your center director or equivalent or delegated authority.

An open source software project release **must exist in a group hierarchy** on https://code.usgs.gov/ (i.e., it may not exist in your personal namespace). Top-level groups should reflect current high-level USGS organizational structure, e.g., science center, mission area, etc... Additional nested groups can be created as appropriate.

An open source software project release may not be cited or referenced by other official USGS Information Products (e.g., manuscripts, data releases). Data produced by an open source software project release may not be used to support any other official USGS Information Product (e.g., manuscript).

Checklist

- Reviews
  - [ ] Administrative security review
- Administrative files
  - [ ] Readme
  - [ ] Disclaimer
  - [ ] License
  - [ ] Metadata
- Request product be made publicly available on a provisional basis
  - [ ] Submit a new issue using the `GitLab Open Source Software Project Release` issue template.

Reviews

Administrative security review

- Ensure review inspects each commit in project history
- If development workflow included incremental peer reviews, this step is already complete

Administrative Files

Readme

There should exist a `README.md` file (casing and extension are important) in the top-level of the project. This file should include introductory information about the project (e.g., title, description, use-case, recommended citation, etc..). It should include links to additional detailed information such as the disclaimer statement, license information, documentation, user guide, related manuscript(s), etc...

This file should be in markdown format to ensure it renders properly on GitLab.

Disclaimer

This FSP Open Source Software Project disclaimer must be included in a file located at the top-level of the project called `DISCLAIMER.md`. Both casing and file extension are important.

This disclaimer may additionally be embedded in the `README.md` file, but this is not necessary and can lead to consistency problems later if the content is updated in once place and not the other. It is instead recommended to link to the disclaimer from the `README.md` file.

License

An appropriate open-source license must be included in a file located at the top-level of the project called `LICENSE.md`. Both casing and file extension are important.

Selecting the appropriate license may depend on the project, is provenance, its dependencies, or other factors. A default starting point is provided, but may not be appropriate in all cases. If you are not sure which license to select, please consult with the Office of the Solicitor for legal guidance.

Metadata

Project metadata are stored maintained in a file at the top-level of the project called `code.json`. This file is in JSON format. Its top-level element should be the releases array as defined in the code.json schema. An example `code.json` template file is provided to help get you started. You may use an online validator tool to check basic syntax and you may validate against the defined schema using a custom command line interface application. Additional information about individual fields is provided below:

- `name`: Should be a short, human readable name for the project. This should match the value provided when creating the project in GitLab.
- `organization`: Must always be `U.S. Geological Survey` casing and punctuation are important.
- `description`: This may be a longer description of the project. It should be nore more than 1-2 sentances. Verbose descriptions may exist in the `README.md` file
- `version`: This should be a semantic version number for the release (e.g., `1.0.0`). This should not include a leading `v` (i.e., `v1.0.0`) or other identifier. Eventually a Git tag should be created for the indicated release.
- `status`: Must be one of the enumerated values listed below:
  - `Ideation`
  - `Development`
  - `Alpha`
  - `Beta`
  - `Release Candidate`
  - `Production`
  - `Archival`
- `permissions.license.name`: If using the default template provided above, this should be `Public Domain, CC0-1.0`. Otherwise it should be the name of the selected license.
- `permissions.license.URL`: A link to the `LICENSE.md` file stored in this project
  - Must use the `raw` variant of the file
  - Must reference the `main` or `master` branch (as appropriate, this will differ for a scientific information product release, see below)
- `homepageURL`: A link to the project homepage
  - Must be publicly accessible
  - Must point to a hosting platform with an authority to operate
    - May point to the project on GitLab

        - May point to a project home page elsewhere on the usgs.gov website
- `downloadURL` : A link to download a ZIP archive of the project source code
    - Must point to the `main` or `master` branch (as appropriate, this will differ for a scientific information product release, see below)
- `disclaimerURL` : A link to the `DISCLAIMER.md` file stored in this project
    - Must use the `raw` variant of the file
    - Must point to the `main` or `master` branch (as appropriate, this will differ for a scientific information product release, see below)
- `repositoryURL` : A link to this project on GitLab
    - Must include the `.git` extension
- `tags` : An array of topical/domain tags relevant to the project
    - If the project leverages or is related to AI/ML in any way, this array must include the tag `usg-artificial-intelligence` . This tag is short for `U.S. Government Aritificial Intelligence` (i.e., do not use usgs-artificial-intelligence).
- `languages` : An array of the programming languages used within this project
- `date.metadataLastUpdated` : An ISO datestamp of when the `code.json` file was last modified. Be sure to update this value whenever you modify anything in this file.

> Additional fields are also available. See the official code.gov metadata schema for additional details.

Note that the top-level element in this file is an *array*. This means it may contain more than one release object for your project; for example, if this project has been under development for a long time, there may be multiple released versions. In this case, release objects should be ordered with the most recently released version appearing first, and so-on in reverse chronological order. For example:

```
[
  {
    // ... release 3.0.0, status Development
  },
  {
    // ... release 2.0.0, status Production
  },
  {
    // ... release 1.0.0, status Archival
  }
]
```

This metadata evolves over time. In the example above, the release tag for version `1.0.0` would only include metadata for that release and it would likely have a status of `Production`. Then in the release tag for version `2.0.0`, two release objects would exist in the array, first would be `2.0.0` with status `Production` and second would appear `1.0.0` with status `Archival`, and so on...

Scientific Information Product Release

The checklist below documents the standard necessary steps/artifacts necessary to complete a scientific information product release of a software product. Individual science centers, offices, etc... may have additional requirements. Check with your local management to determine if any additional steps apply.

Scientific information product releases **must** refrence immutable Git tags. As you work towards a scientific information product release, you can use a `release candidate` branch to facilitate this process. The release candidate branch should have the same name as the eventual release tag. Upon completing the scientific information product release process, all other branches, tags, refs, etc... in the associated GitLab project are automatically released as provisional products as well. The primary distinguisher between an open source software project release and a scientific information product release are the contents of the `DISCLAIMER.md` file, the `code.json status` property, and the ability to cite/reference the released product.

Checklist

- ☐ Create IPDS record
  - Document reviews and approvals in this record as you proceed
- ☐ Reserve a DOI
  - This should point to the immutable tag to be created later, it will not resolve until the tag is created, that is okay
- ☐ Complete checklist for an open source software project release (above)
  - It is acceptable to skip submitting the issue to request the open source software project release when directly pursuing a scientific information product release
- ☐ Create a release-candidate branch for the targeted release
  - If intending to release an immutable tag version `1.0.0`, create a *release candidate* branch `1.0.0`.
- Reviews
  - ☐ Technical code review

        - Should be completed by somebody with adequate understanding of the programming languages used in the project
        - May be facilitated/reconciled via GitLab issues and merge requests against the release candidate branch
        - Must be documented in the IPDS record
    - ☐ Domain Review
        - Should be completed by somebody with adequate understanding of the scientific subject matter covered in the released product
        - Solid documentation and testing suites can aid this review process
        - Must be documented in the IPDS record
- Reconcile `main` or `master` branch
    - ☐ Merge changes from the release candidate branch back into the `main` or `master` branch (as appropriate)
- Prepare final release changes (in the release candidate branch)
    - ☐ Update `DISCLAIMER.md` file to include the official verbiage
    - ☐ Update the `code.json` file...
        - The `version` field should be updated appropriately
        - The `status` field can be updated appropriately
        - URLs should point to the immutable tagged version (i.e., not `main` or `master`)
    - Do **not** merge these changes back in to the `main` or `master` branch
- Request product be made publicly available as an official USGS Information Product
    - ☐ Submit a new issue using the `GitLab Scientific Information Product Release` issue template.
    - The `Branch or Tag` should indicate the intended target release tag even though it does not exist yet. The corresponding release candidate branch will be used to finalize the release.
    - An administrator will contact you to correct any errors in the release package and facilitate the final release of the product, during this process, you will be instructed to create an immutable Git tag and delete the release candidate branch.

References

1. https://code.chs.usgs.gov/software/software-management/-/issues/new
2. https://www.usgs.gov/software-management/types-software-review
3. https://docs.gitlab.com/ee/user/markdown.html

4. https://code.chs.usgs.gov/software/software-management/-/raw/main/administrative_templates/DISCLAIMER_preliminary.md
5. https://code.chs.usgs.gov/software/software-management/-/raw/main/administrative_templates/LICENSE.md
6. https://www.json.org/json-en.html
7. https://github.com/GSA/code-gov-data/blob/master/schemas/schema-2.0.0.json
8. https://code.chs.usgs.gov/software/software-management/-/raw/main/administrative_templates/code.json
9. https://jsonformatter.curiousconcept.com/
10. https://code.chs.usgs.gov/ghsc/hazdev/inventory-validator
11. https://semver.org/
12. https://git-scm.com/book/en/v2/Git-Basics-Tagging
13. https://code.chs.usgs.gov/software/software-management/-/raw/main/administrative_templates/DISCLAIMER_official.md